Dongwook Kwon

+82 10-6612-9246 | dwkwon@suresofttech.com | dongwooky.github.io | github.com/dongwooky | linkedin.com/in/dongwooky

Summary

Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization, Multi-Agent Systems (MAS), and AI-driven anomaly detection for critical infrastructure. Demonstrated research excellence through co-first authored publications in top-tier peer-reviewed venues, including an Oral presentation at EMNLP 2025 and a paper at ACL 2026 (both Industry Tracks), developed in collaboration with LG Electronics. Inventor of multiple registered patents in network and medical device anomaly detection. A core engineer who engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company. With foundational research experience in nuclear cybersecurity and medical cyber-physical systems, poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

Patents

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

Publications

Journal Articles

Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

D. Kwon, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

Conference Papers

R. Chang†, **D. Kwon†**, J. Lee†, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. †Equal contribution.

J. Lee†, R. Chang†, **D. Kwon†**, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. †Equal contribution.

D. Kwon, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

D. Kwon, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

Research & Professional Experience

AI Engineer, Intelligent Agent Technology Team — Dec. 2025 – Present
Suresoft Technologies — *Seongnam, Korea*

- Developing an AI validation system utilizing AI agent–based architectures for enterprise model verification.
- Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.
- Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.

Research Assistant — Aug. 2021 – Dec. 2025
Bio-Computing & ML Lab, Kwangwoon University — *Seoul, Korea*

- Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.

- Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.
- Published in EMNLP 2025 (Oral) and ACL 2026; co-author on two granted Republic of Korea patents.

Visiting Researcher Jan. 2025 – Jul. 2025
LG Electronics Canada *Toronto, ON, Canada*

- Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.
- Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.

AI Development Project Participant, Summer Program Jul. 2022 – Aug. 2022
Qualcomm Institute, UC San Diego *San Diego, CA, USA*

- Built a personality-based drug-addiction prediction system using machine-learning models.
- Recognized with the program's Achievement Award.

Education

Kwangwoon University Seoul, Korea
M.S. in Computer Engineering, GPA 4.33 / 4.5 (98.1%) *Mar. 2024 – Feb. 2026*

University of Toronto Toronto, ON, Canada
Graduate Research Program, GPA 3.68 / 4.0 *Jan. 2025 – Jul. 2025*

Kwangwoon University Seoul, Korea
B.S. in Electronics and Communications Engineering, GPA 3.41 / 4.5 (88.1%) *Mar. 2019 – Feb. 2024*

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project — UC San Diego

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024)

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering